How Far Does a Litter Detector Travel?

Cross-Domain Urban Waste Detection Across Handheld, Dashcam and Drone Imagery

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The question

Litter detection benchmarks are evaluated **in-domain**, yet any city deploying mixed sensors — citizen photos, garbage-truck dashcams, drones — implicitly assumes detectors transfer between viewpoints. **Nobody had measured whether they do.** We train and evaluate across three public domains under one audited protocol:

Dataset	Viewpoint	Images	Median obj.	% small
TACO	handheld	1,500	171 px	25
RoLID-11K	dashcam	11,564	**22 px**	84
UAVVaste	drone	772	72 px	66

19 experiments, 4 model families (Faster R-CNN, YOLOv11n, FCN, VGG+CAM), every metric from the same pycocotools contract against frozen ground truths — tables and figures are generated, never transcribed.

Auditing before training: 7 defects found

Systematic checks (perceptual hashing, EXIF verification, split cross-referencing) on the released datasets surfaced:

- **H4 — video-level leakage in RoLID-11K's official splits:** 58.2% of test frames share a source video with training (87% intra-video visual redundancy). **Published baselines are inflated by scene memorization.**
- H1/H2 — TACO basename collisions across batches; ∼37% pending EXIF rotations with boxes annotated on the rotated frame.
- H5/H6 — duplicate image between val and test (RoLID); near-duplicate pair crossing train/test (UAVVaste).
- H3/H7 — release folders do not encode capture hour; annotation frames inconsistent across datasets.

Cure: medallion pipeline (raw→bronze→silver →gold) with SHA-256 lineage, five corrections, and frozen group-aware splits (pHash groups / source video; iterative stratification puts every class within 0.1 pp of the 15% test target).

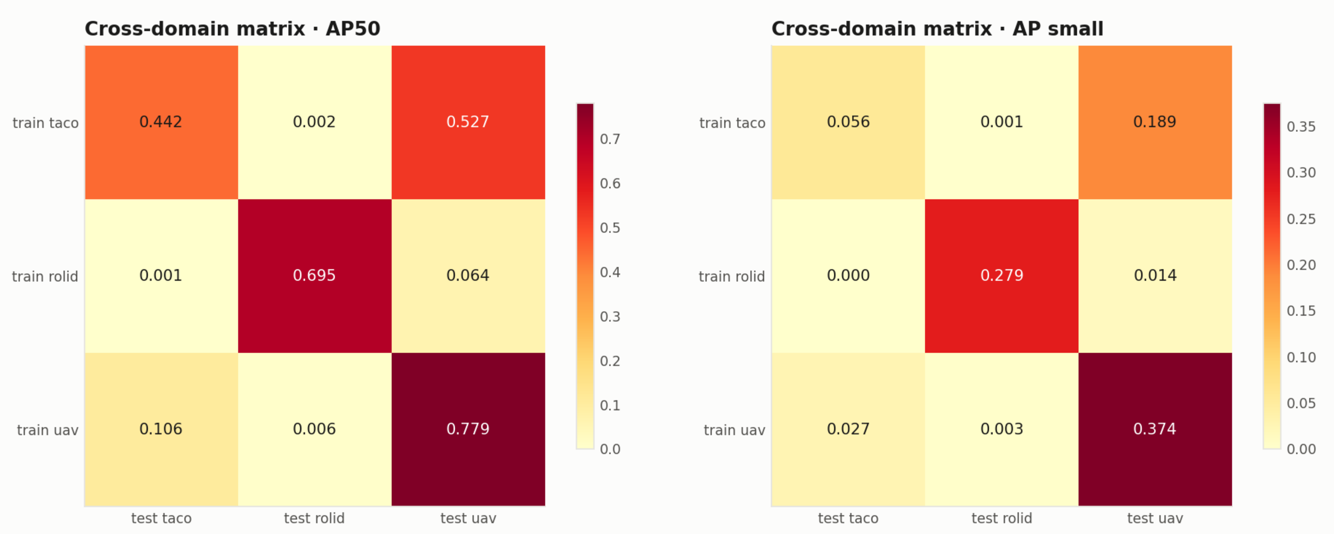
82%
cross-domain mAP_{50} drop

+21%
AP_{50} inflation from split leakage

+8.3
AP_{50} from 640→1024 px (dashcam)

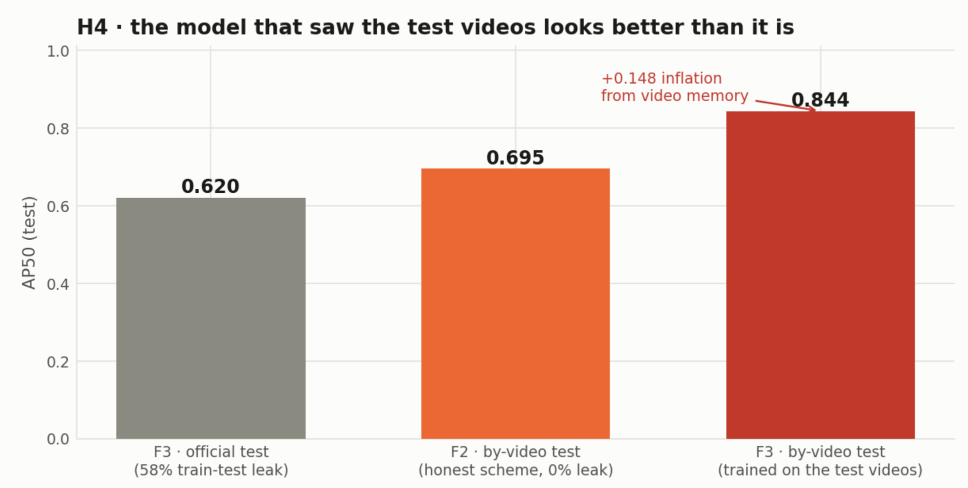
16×
smaller model, within 1–4 AP_{50} points

Detectors do not travel



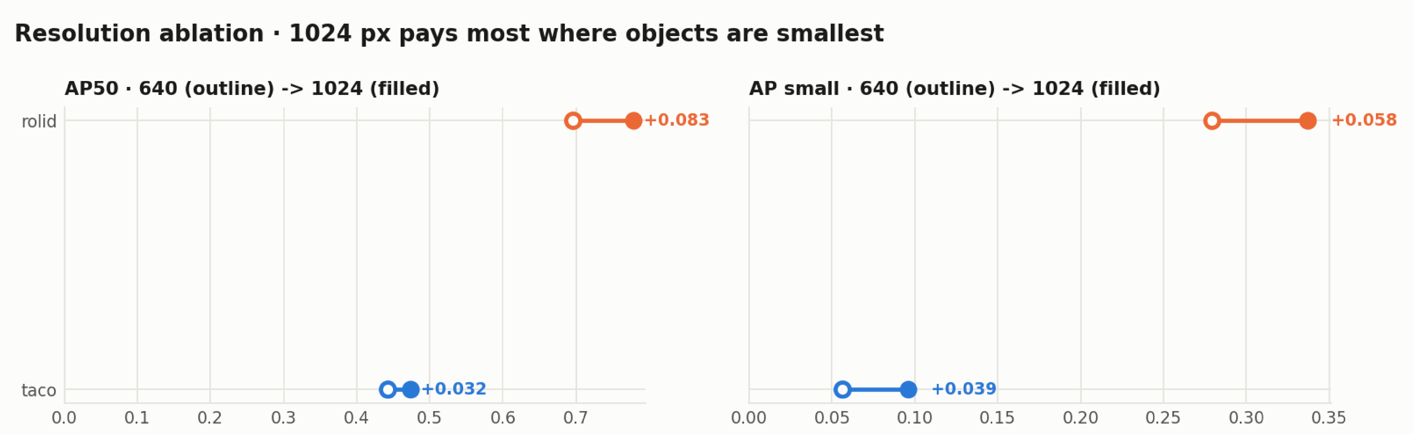
In-domain mean 0.639 AP_{50}; cross-domain 0.118. The dashcam domain is an **island** ($\mathrm{AP}_{50} \leq 0.064$ both directions): its 22-px median object falls below the smallest 32-px FPN anchor. Handheld→drone retains 68% — scale, not appearance, dominates the gap.

Split hygiene changes conclusions



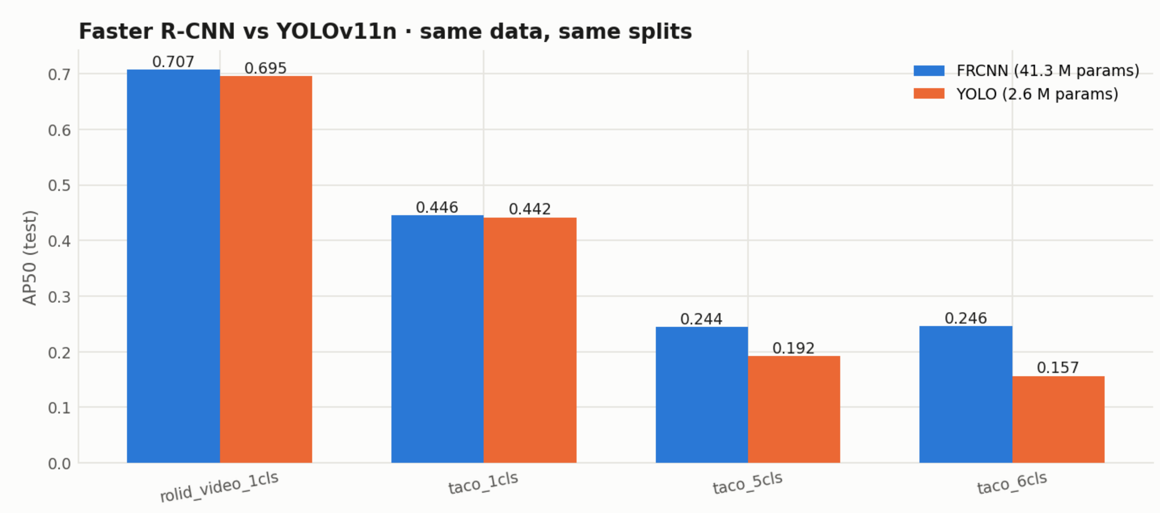
On the *same* clean test set, the model trained on the leaky official split scores **0.844 vs. 0.695** for the cleanly trained one: **+21% relative** — memorized scenes, not generalization. Group-aware (by-video) splitting removes the bias at zero cost.

Cheap levers: resolution



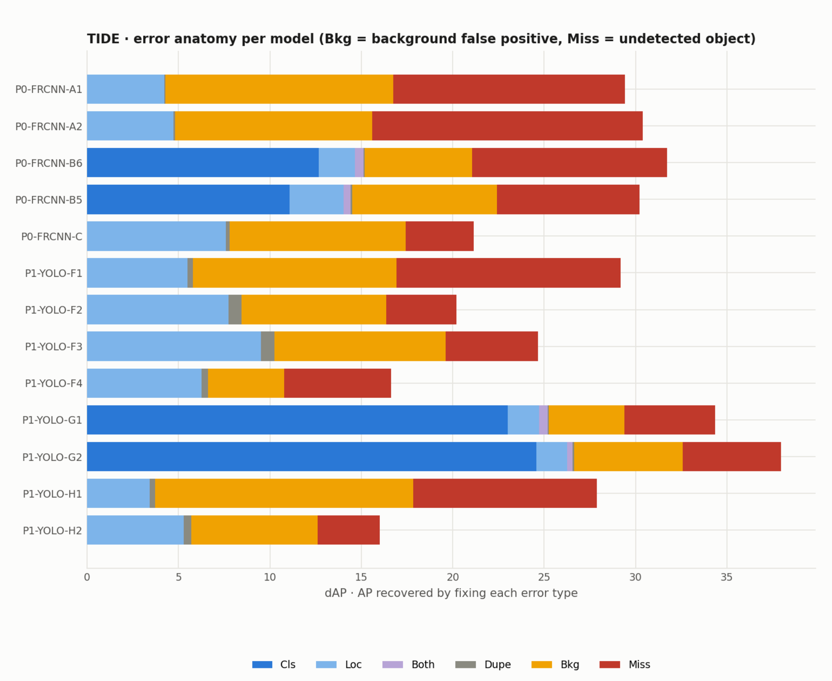
Gains concentrate where objects are smallest: dashcam +8.3 AP_{50} / +5.8 AP_S; handheld +3.2 / +3.9. Class granularity: excluding the marginal glass class (38 test instances, AP_{50} 0.049) lifts the remaining five classes by +0.014 AP_{50} on average.

41M vs 2.6M parameters, same splits



YOLOv11n stays within 1–4 AP_{50} points of Faster R-CNN on single-class litter (0.695 vs 0.707 dashcam) — lightweight deployment is viable. The gap widens on materials (0.157–0.192 vs 0.244–0.246). FCN: 0.577 test IoU; VGG-16 crops: 97.8% accuracy with CAM evidence on the object.

Why they fail: TIDE error anatomy



Multiclass models lose up to **24.6 dAP to classification** vs. 1.7 to localization — naming the material, not finding the litter, is the bottleneck; in-domain dashcam models miss little ($\mathrm{dAP}_{\mathrm{Miss}} \leq 3.9$).

Takeaways & reproducibility

- **No universal litter model yet**: per-domain fine-tuning (or at least resolution adaptation) remains mandatory for mixed-sensor deployments.
- **Audit before you train**: one hidden leak rewrites a benchmark; video/burst datasets should ship group identifiers.
- **Detect class-agnostic, classify crops**: TIDE argues for decoupling material recognition (97.8% on crops) from detection.

Three released routes: demo with pretrained weights on any photo (∼5 min, CPU) · regenerate every paper number from artifacts (∼10 min) · full pipeline from public data. Code, weights, splits and video: **github.com — released with the course hand-in.**